Jeffrey Murray Jr

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TECHNICAL SKILLS

Languages: Python, Clojure, C++, Dart, TypeScript, Rust

Infrastructure: Kubernetes, Docker, Helm, AWS

Focus: Low-Latency Distributed Computing, Real-Time Communication, Privacy & Security

EDUCATION

Master of Science Cybersecurity Engineering *2024*
University of Washington | GPA 3.83

Bachelor of Science Computer Science and Software Engineering *2020*
University of Washington | GPA 3.56

PROFESSIONAL EXPERIENCE

Senior Developer *Jan 2022 - Present*
SAP Concur – Bellevue, WA

- Architect low-latency microservices handling 500M+ API requests/day at sub-100ms p95.
- Delivered RFC 7644 Filtering to search across 400M+ documents at sub-second p95 at 5M+ requests/day.
- Launched a distributed load testing platform on Kubernetes simulating 10K+ concurrent connections.
- Built internal MCP server and Claude Skill to accelerate API onboarding for downstream teams.

Research Assistant *Jul 2020 – Sep 2020*
University of Washington – Bothell, WA

- Privacy Analysis of Data Anonymization of Customer Proprietary Network Information.
- Delivered two workshop presentations and a whitepaper for our telecommunications partner.

Software Developer Internship *Oct 2019 – Feb 2020*
Apollo Video Technology – Bothell, WA

- Developed a Flutter mobile application from scratch to decode live video streams.
- Rendered live video streams over WebSockets using Flutter's WebView plugin.
- Designed layered architecture for user interface handling asynchronous network calls in Dart.
- Collaborated with three team members to design and QA test application.

TECHNICAL PROJECTS

Toward Easier Development of Privacy-Preserving Mobile Crowdsensing Applications *Jan 2024 – Dec 2024*
Fully Homomorphic Encryption

- Engineered GhostPeerShare for privacy-preserving comparison of encrypted video similarity.
- Abstracted Microsoft SEAL library for Dart plugin to manage C objects by reference within the memory stack.
- Facilitated proximity-based data exchange via Bluetooth and Wi-Fi on Android.
- Led a team of three through the full application lifecycle: design, implementation, and QA testing.

Smart Mirror with Facial Recognition *Sep 2021 – Dec 2021*
Internet of Things, Machine Learning

- Delivered voice-controlled interface via Google Assistant and Porcupine wake-word detection.
- Extended Magic Mirror platform in Node.js as a user interface on Raspberry Pi 4.
- Use of CUDA powered facial recognition in Python on Jetson Nano.
- Created RESTful API on Google Cloud Services to facilitate event-driven communication.

PUBLICATIONS

J. Murray and B. Lagesse, "Toward Easier Development of Privacy-Preserving Mobile Crowdsensing Applications," IEEE International Conference on Mobile Data Management (MDM), 2025, pp. 264-269.

J. Murray, A. Mashhadi, B. Lagesse, and M. Stiber, "Privacy Preserving Techniques Applied to CPNI Data: Analysis and Recommendations," arXiv preprint arXiv:2101.09834, 2021.

A. Mashhadi, J. Sterner, and J. Murray, "Deep Embedded Clustering of Urban Communities Using Federated Learning," IEEE International Joint Conference on Neural Networks (IJCNN), 2021, pp. 1-8.